

Some Popular Handheld Multimeters

	Fluke 175	Agilent U1232A	BK Precision 2709B	Motwane M63	Meco 450 TRMS	Metrix + 18xL
						
Range						
Voltage	600 mV-1000V	600 mV-600V	660 mV-1000V	600 mV/1000V	200 mV-1000V	600 mV-1000V
Current	60 mA-10A	60 μ A-10A	660 μ A-10A	60 mA/20A	20 mA-10A	600 μ A-600A
Resistance	600 Ω -50 M Ω	600 Ω -60 M Ω	660 Ω -66 M Ω	600 Ω -60 M Ω	200 Ω -20 M Ω	600 Ω -60 M Ω
Diode	2.4V	2V	3.5V		8V	2.8V
Capacitance	100 nF-9999 μ F	1000 nF-10 mF	6.6 nF-66 mF	40 nF - 4000 μ F	20 nF-20 μ F	40 nF-400 μ F
Frequency	99.99 kHz	99.99 kHz	66 MHz	10 MHz	200 kHz	10 MHz
Temperature	-40 to 400°C	-40 to 1372°C	NA	NA	NA	-50 to 700°C
Resolution	6000 counts	6000 counts	6600 counts	6000 counts	19,999 counts	6000 counts
DC accuracy	0.15% + 2 counts	0.5% + 2 counts	0.5% rdg + 2 digits	0.5% + 3 counts	0.05% rdg + 3 digits	0.8% rdg + 10 digits
Input impedance	10 M-ohms	10 M-ohms	10 M-ohms	10 M-ohms	10 M-ohms	10 M-ohms
True RMS	Yes	Yes	Yes	Yes	Yes	Yes
Display updates	4/sec	5/sec	2/sec	3/sec	2.5/sec	3/sec
Autoranging	Yes	Yes	Yes	Yes	No	Yes
Capacitance meter	Yes	Yes	Yes	Yes	Yes	Yes
Frequency meter	Yes	Yes	Yes	Yes	Yes	Yes
Temperature measurement	Yes	Yes	No	No	No	Yes
Battery life	400 hrs	500 hrs	150 hrs	200 hrs	150 hrs	120 hrs
Power consumption		450 mVA maximum		4 mA		8W (approx.)
CAT3	Yes	Yes	Yes	CAT IV/600V	CAT1 (1000V max.)	Yes
Fuse	11A, 1000V fast-acting fuse	10 $\times$ 38mm 11A/1000V fast-acting fuse		20A/500V fast blow type ceramic fuse		6 $\times$ 25mm 10A/250V

capacitance, frequency and temperature also. For regular electronics jobs, the resistance range should be at least 20 megaohms. In case of capacitance measurement, the smaller the minimum range, the better. As very high values of capacitance are not very relevant in electronics, focus on the minimum range here.

Frequency measurement is a little overrated feature as nobody will measure frequency using a multimeter. Also, the range of frequencies supported by a multimeter is very small (say, 100 kHz). You would need to measure much higher frequencies when working with electronic circuits and a multimeter cannot be used in that case.

Temperature measurement is a handy feature. Normally, multimeters with temperature measurement function are provided with a K-type thermocouple cable to measure temperature. Check the minimum and maximum range of temperature measurement in the datasheet and see if that meets your requirement.

Another range that you need to check is the diode function range. Choose a multimeter with a good range. For example, if the diode function range is only 2V, you will not be able to test an LED with more than 2V forward voltage drop.

Resolution. Resolution of a multimeter is the smallest reading that can be shown on the display. Earlier

described in digits like 4½ and 5½, these days resolution is specified in counts.

To help you understand resolution, let me take the example of a 4.5-digit or 4½-digit multimeter. Here '4' means four full digits that can take up values from '0' to '9' on the display. Fractional digit represents the maximum value that the first digit (most significant digit) can display. In this case, '1/2' or '0.5' means that the first digit can have values '0' or '1'. Therefore 4½-digit means that the display can show values from 00000 to 19999, which totals 20,000 counts.

To calculate the smallest reading that can be shown on the display, just divide the full-scale reading by the